

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – 3.2 LINEAR MOTION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the Driving Question: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes? Work on your own. Write your answers in the space under each prompt. Use what you observed, measured and calculated in Lessons 1 to 7. Show every calculation: write the known values with units, name the equation you choose, and give your final answer with units. Use $g = 10 \text{ m/s}^2$ in all calculations. Ignore friction and air resistance unless the question says otherwise. Remember Mzee Kamau's mkokoteni (300 kg when loaded) is released from rest on a straight, steady slope on the Limuru-Nairobi escarpment road where $\sin(\text{angle}) = 0.2$. It crashes into a parked lorry after 100 m along the road.

Part 1: Explaining what Achieng saw

Achieng, a bystander, said: 'It started slowly then flew.' Mzee Kamau's mkokoteni has no engine and nobody was pushing it. Write a Claim, Evidence, Reasoning explanation of why the cart keeps getting faster. In your answer:

- (a) State your claim.
- (b) Use at least three values from the cart's motion (for example, its speed at different times) as evidence.
- (c) Explain why the cart started slowly, and why the crash felt so sudden, using the idea of how distance changes with time.
- (d) Say whether the cart's motion is free fall, and explain your answer.

Write your answer here:

Part 2: Predicting the crash using the equations of linear motion

The cart is released from rest ($u = 0$) with $a = 2.0 \text{ m/s}^2$ and crashes after travelling 100 m along the slope.

- (a) List the five variables used in the equations of linear motion and the knowns and the unknown for finding the speed at the crash.
- (b) Choose one of the four equations of linear motion, explain in words why it is the right one, and calculate the speed at the crash in m/s and in km/h.
- (c) Use a different equation to find the time taken to reach the lorry, and show that it agrees with the data from the lessons.
- (d) Find the speed of the cart after it has travelled 25 m. Then compare 25 m with 100 m: by what factor has the distance changed, by what factor has the speed changed, and what does this tell you about how speed depends on distance?

Write your answer here:

Part 3: Free fall compared with the cart

Class drop data: a rubber bung, a stone and a crumpled paper ball were each dropped from $h = 2.0$ m. The mean times of 5 trials were 0.63 s, 0.64 s and 0.65 s respectively.

(a) What do these results show about the effect of mass on how objects fall when air resistance is negligible?

(b) Use the class mean time of 0.64 s to estimate g , and explain why we use $g = 10 \text{ m/s}^2$ in calculations.

(c) A stone is dropped from rest through a vertical drop of 20 m, the same vertical drop as the cart's. Calculate the time taken and the speed on reaching the ground.

(d) Compare your answers in (c) with the cart's time and speed. Explain the difference in time using the idea of the slope, and say what type of slope free fall represents.

Write your answer here:

Part 4: Slowing down and the sign convention

A boda-boda rider, Otieno, is travelling at 15 m/s and brakes before a police check with a constant acceleration of -3 m/s^2 .

(a) Explain what the negative sign tells you, and compare it with the sign of the cart's acceleration down the slope.

(b) Calculate how long Otieno takes to stop and how far he travels while stopping.

(c) A ball is thrown upward at 15 m/s from a 20 m high Mombasa rooftop. Taking up as positive, state the acceleration of the ball at the top of its path and explain your answer. Then calculate its velocity when it hits the ground.

Write your answer here:

Part 5: Road safety and the final explanation

A matatu on Mombasa Road has a driver reaction time of 1.0 s. On a dry road, braking gives a deceleration of 5 m/s^2 . On a wet road, the deceleration is 2.5 m/s^2 .

(a) The matatu travels at 72 km/h. Convert this to m/s, then calculate the reaction distance, the braking distance on a dry road and the total stopping distance.

(b) Find the braking distance of the same matatu on a wet road, and compare it with the dry road.

(c) The matatu's speed is increased to 108 km/h. Find the new braking distance on a dry road, and say by what factor it has changed compared with part (a) and why.

(d) Explain how your answers help us understand why the Kinangop descent has warning signs and why a runaway mkokoteni with no brakes is so dangerous. Suggest TWO ways to reduce the dangers of speed on steep roads.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the phenomenon (claim, evidence, reasoning)	Gives a clear claim that gravity acts down the slope all the time giving a constant acceleration of 2.0 m/s^2 . Uses at least three correct cart values (such as 4 m/s at 2 s, 10 m/s at 5 s, 20 m/s at 10 s and 100 m). Explains why it starts slowly and why the crash feels sudden ($s = t^2$), and correctly says the cart is not in free fall.	Gives a correct claim with some correct data, and explains acceleration, but explains only one of 'starts slowly' or 'feels sudden', or does not address free fall.	Claim is unclear or only says the cart is 'pulled' or 'pushed' without linking to constant acceleration, with little or no data used.

Using the equations of linear motion	Lists the five variables with units, writes the knowns and unknown, chooses a suitable equation and justifies the choice, calculates correctly with units (20 m/s, 72 km/h, 10 s, 10 m/s at 25 m), verifies with a second equation, and correctly compares how distance and speed change between 25 m and 100 m.	Chooses suitable equations and gets most answers correct with units, but gives little justification for the choice or does not check with a second equation.	Equations are chosen at random or substituted incorrectly, or answers have missing or wrong units.
Free fall and the comparison with the slope	Explains that mass does not change the fall time, estimates g correctly (9.8 m/s^2) and explains the use of 10 m/s^2 . Calculates 2 s and 20 m/s for the 20 m drop and explains that the slope gives only 0.2 of g , with free fall as the extreme case of a vertical slope.	Calculates the free-fall values correctly and says the cart is slower, but the explanation of why (the slope fraction of g) is incomplete.	Does not link results to the effect of mass or to the slope, or uses $a = 10 \text{ m/s}^2$ for the cart.
Sign convention and use of scalar/vector terms	Chooses a positive direction explicitly, interprets negative acceleration as opposite to the motion, correctly uses $a = -10 \text{ m/s}^2$ for the upward-thrown ball including at the top, and gets the 37.5 m, 5 s and -25 m/s answers with correct signs.	Uses signs mostly correctly and gets most answers right, but does not explain the signs or makes one sign error.	Does not use signs consistently, or says the acceleration is zero at the top of the ball's path.
Applying the ideas to road safety	Converts units correctly, calculates reaction, braking and total stopping distances (20 m, 40 m, 60 m; 80 m wet; 90 m at 30 m/s), explains the square relationship, and gives two sensible ways to reduce danger linked to Kinangop signs and NTSA rules.	Calculates most distances correctly and gives sensible safety measures, but explanation of the square relationship is weak.	Makes errors in unit conversion or stopping distance, or gives safety ideas that are not linked to the calculations.

ATTRIBUTION AND LICENSING

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